

Energy use and environmental impact of three lithium-ion battery factories with a total annual capacity of 100 GWh

Ákos Kuki, Csilla Lakatos, Lajos Nagy, Tibor Nagy, Sándor Kéki*

Department of Applied Chemistry, Faculty of Science and Technology, University of Debrecen, Egyetem tér 1, H-4032 Debrecen, Hungary

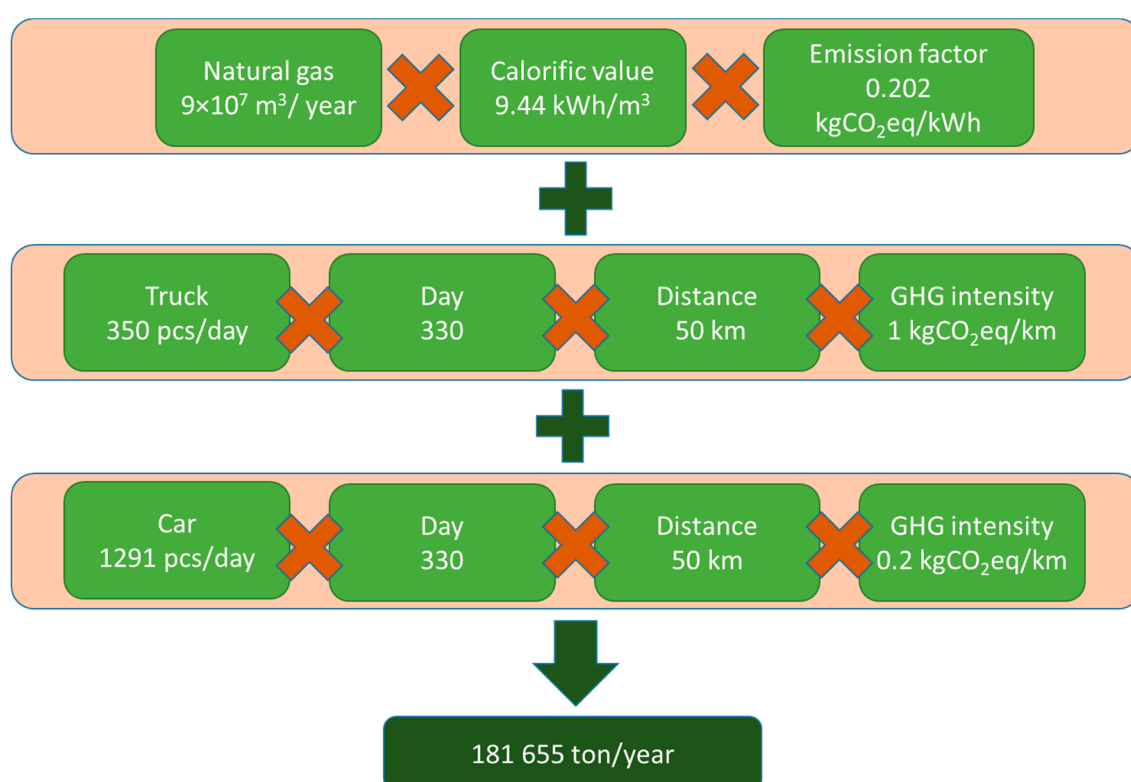


Figure S1a. Scheme of calculation of *local* greenhouse gas emissions of the CATL plant in Debrecen, Hungary

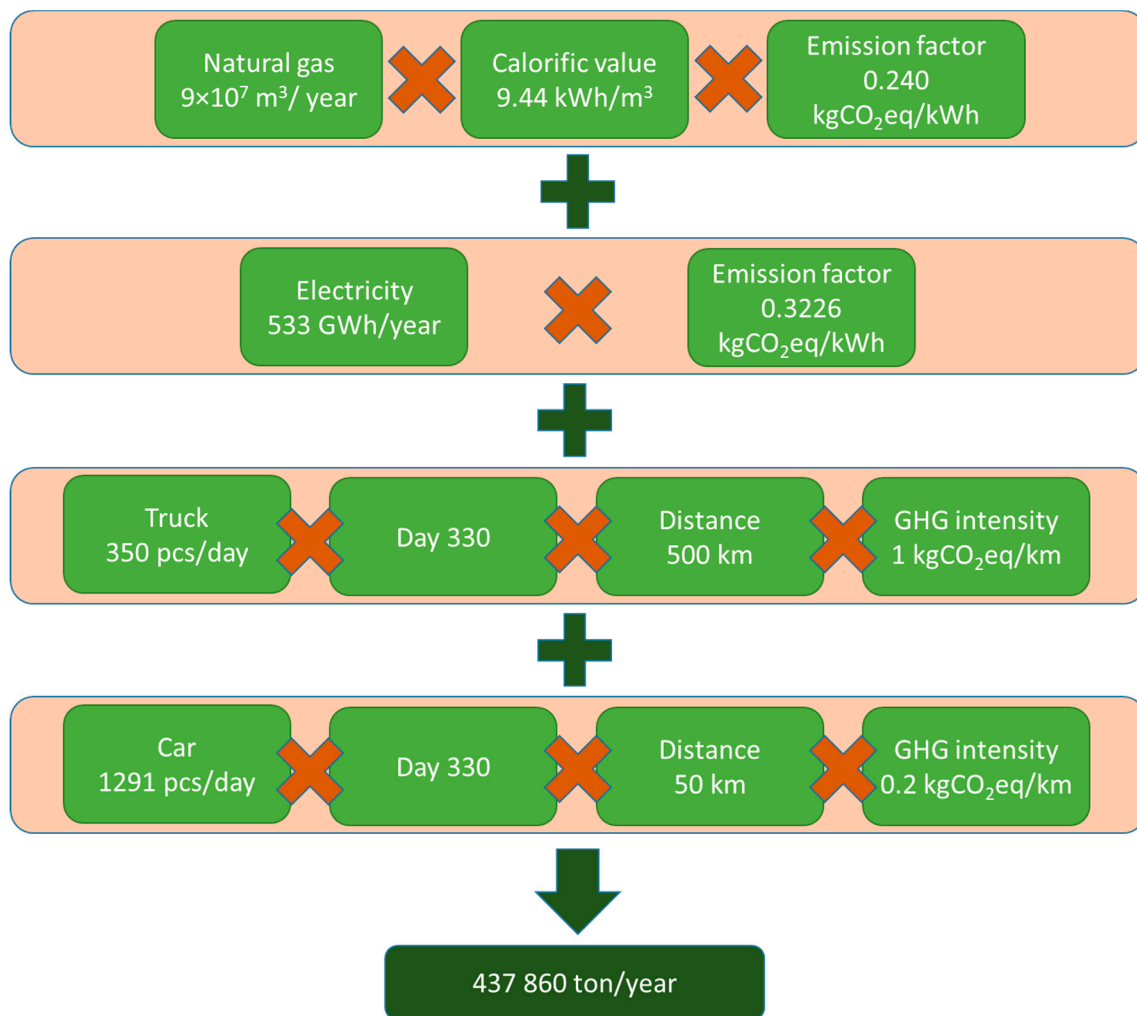


Figure S1b. Scheme of calculation of *global* greenhouse gas emissions of the CATL plant in Debrecen, Hungary

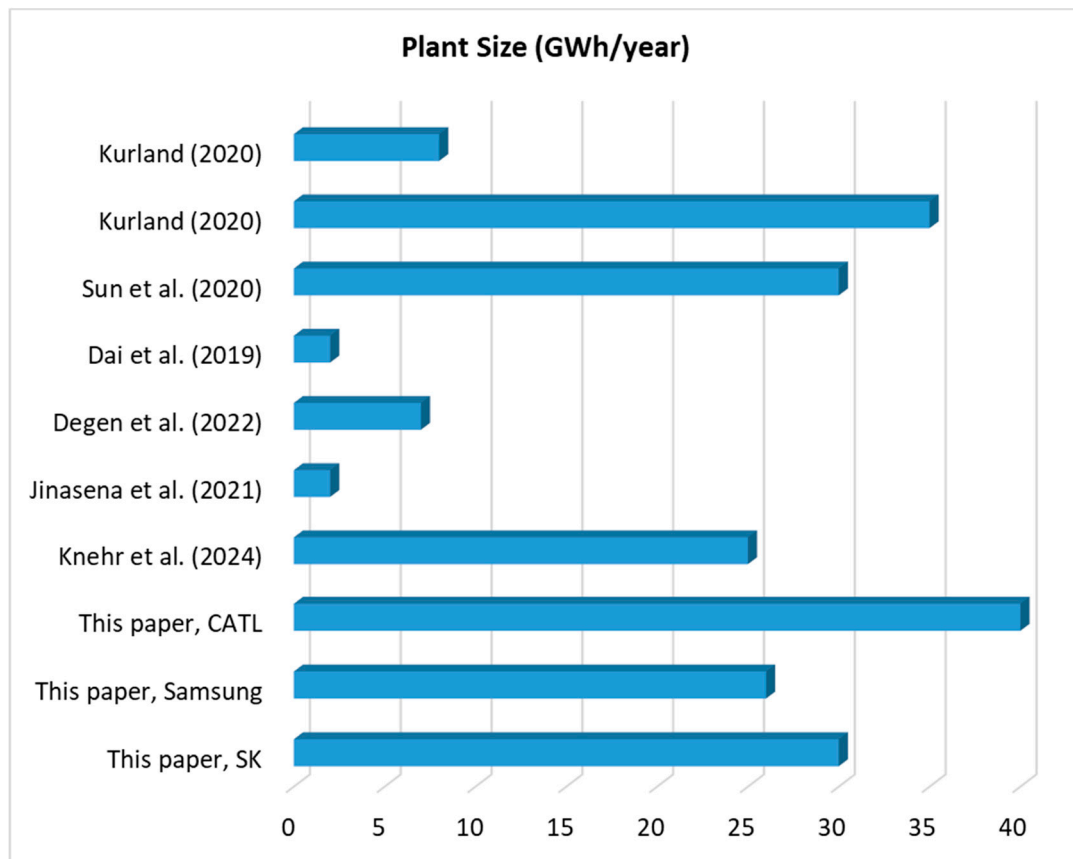


Figure S2a. Comparing plant size with previous studies

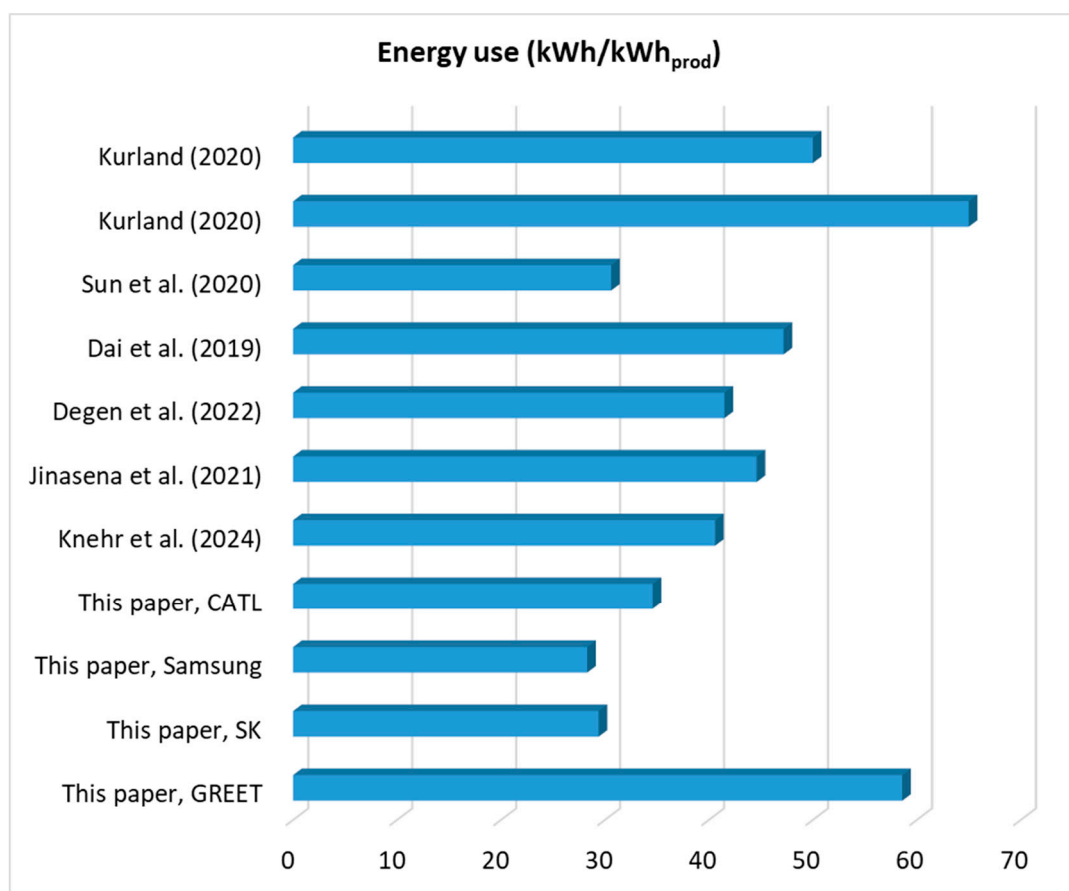


Figure S2b. Comparing energy use with previous studies

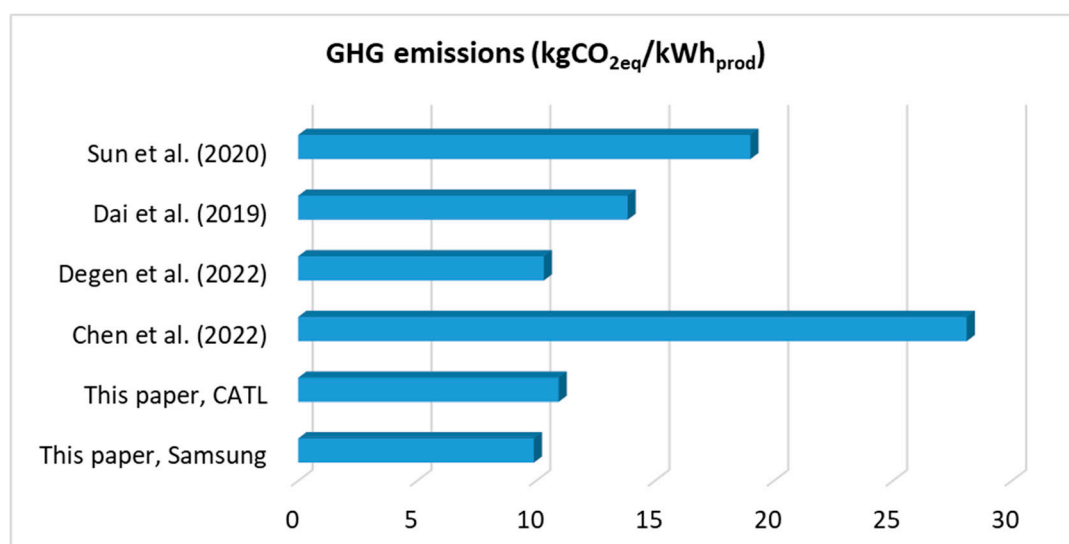


Figure S2c. Comparing GHG emissions with previous studies

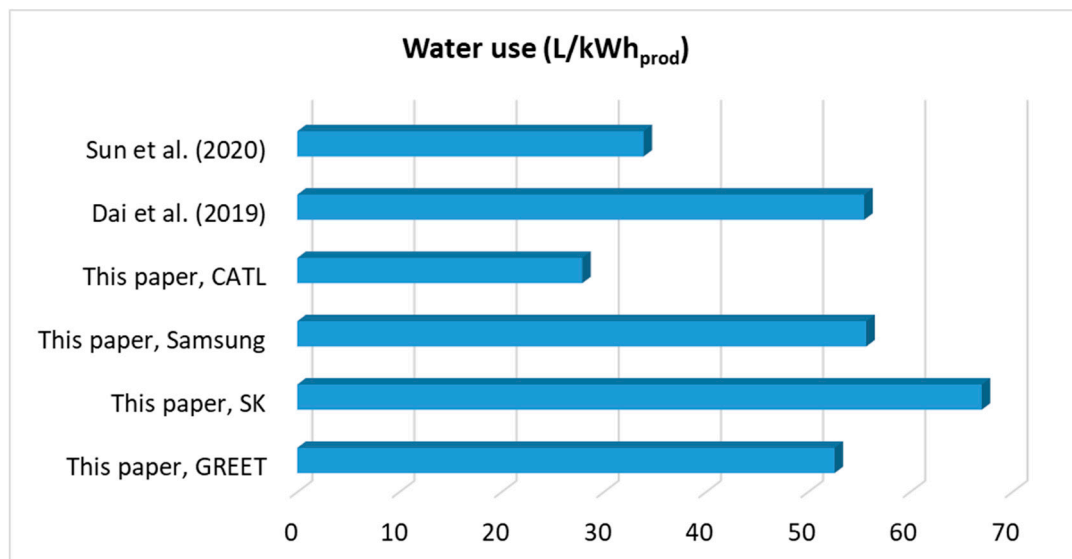


Figure S2d. Comparing water use with previous studies

English translations of the relevant parts of the references in Hungarian (ref. 16-19).

The data used in the article is highlighted in red.

[16] Contemporary Amperex Technology Hungary Ltd. Battery manufacturing plant Modification of the Uniform Environmental Permit, Complete environmental review (2024)

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Table 5. Basic data on the operation

<i>Designation</i>	<i>Own location</i>	<i>Hired location</i>
Battery manufacturing capacity (GWh)	40	
Battery assembly capacity (GWh)		9.2
Combustion equipment (MWh)	175	
Average water demand (m ³ /day)	3 378	34.05
Peak water demand (m ³ /day)	6 232	414
Evaporation loss	85%	17%
...		
Traffic:		
Trucks (pcs/day)	350	45
Cars (pcs/day)	1291	185

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Used energy:

- Natural gas: 90 000 thousand m³/year
- Electricity: 533 GWh (Purchased from the electricity grid)

[17] Samsung SDI Hungary Plc., Göd, Uniform Environmental Permit, 2023.

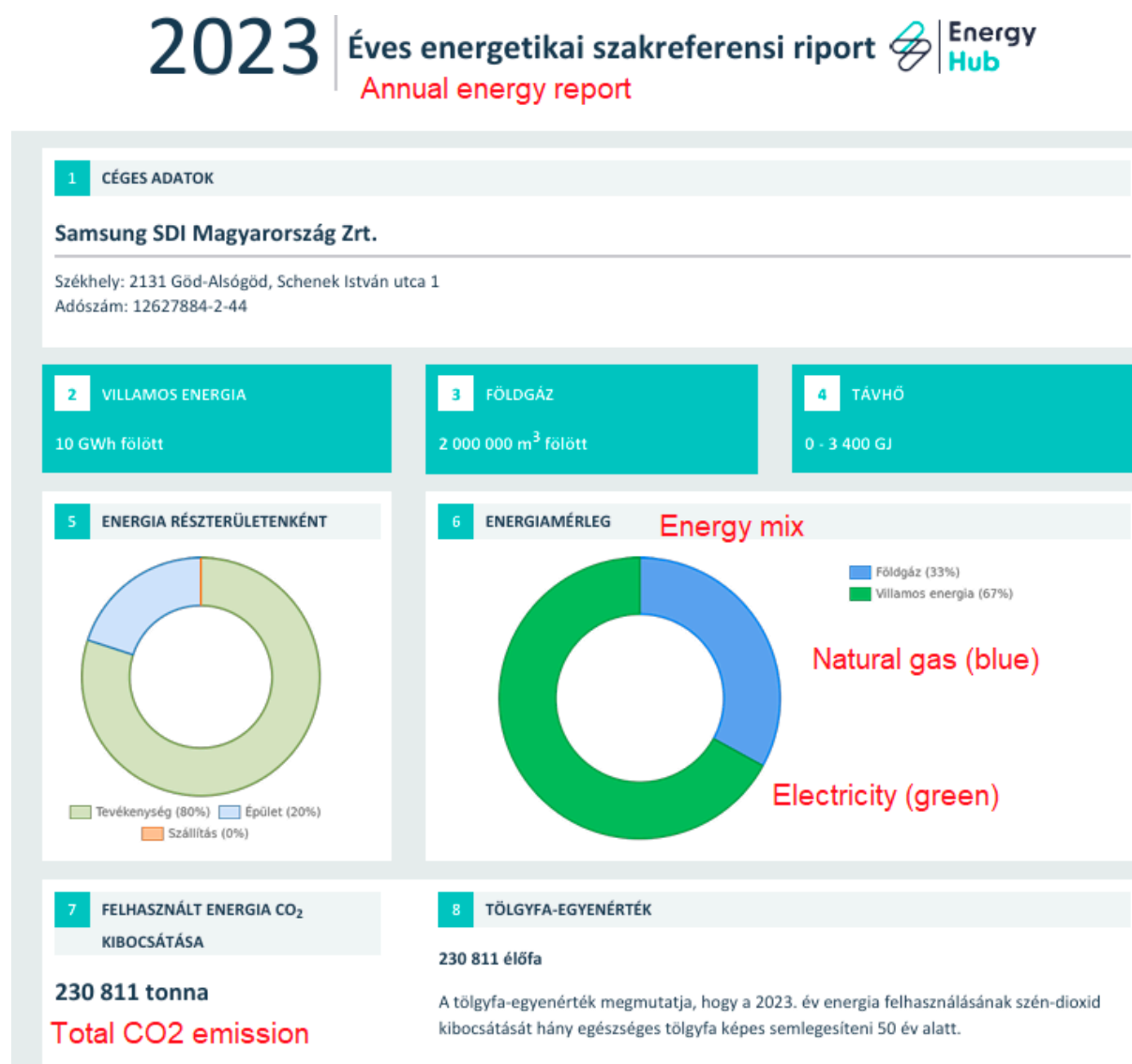
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Battery manufacturing capacity planned in 2023: 130 900 ton/year

How much further increase in **water consumption** can be expected, as the study shows that 71% of the total consumption in 2022 (1,458,927 m³) will have been consumed in the first half of 2023 (1,037,656 m³)?

[18] Annual energy report, 2023

https://samsungsdi.hu/upload/eves_szakreferensi_riport_2023_hu.pdf.



[19] SK ON Hungary Ltd., Uniform Environmental Permit, 2023.

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3.9 Material and energy use, traffic:

Battery manufacturing capacity: 30 GWh/year

Input material and energy streams:

Supply of raw materials and consumables: 150,300 ton/year

Natural gas: 11,817 m³/h (for the combustion equipment)

Water use total: 5,200-6,982 m³/day, of which 576 m³/day is communal, 4,624-6,406 m³/day is industrial

Total NMP use (standard operating max.): 13,150 ton/year, (17,000 ton/year design max.)

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Traffic:

Vehicle	Direction	Time	Pcs
Truck	In	Day 6:00-22:00	95
		Night 22:00-6:00	30
	Out	Day 6:00-22:00	95
		Night 22:00-6:00	30
Bus	In	Day 6:00-22:00	45
		Night 22:00-6:00	20
	Out	Day 6:00-22:00	42
		Night 22:00-6:00	20
Car	In		300
	Out		300